House Price Prediction using Linear Regression

Aim : To build a Linear Regression model to predict house prices based on square footage, number of bedrooms, and bathrooms.

Objective : The objective of this project is to analyze housing data and develop a machine learning model that can accurately predict house prices using selected features.

Dataset Description : The dataset used is the House Prices dataset. It contains information about houses such as area (GrLivArea), number of bedrooms, number of bathrooms, and sale price.

Tools & Technologies : Python, Pandas, NumPy, Matplotlib, Scikit-learn, Google Colab.

Procedure :

1.Imported required libraries

2.Loaded the dataset

3.Selected important features

4.Handled missing values

5.Split data into training and testing sets

6.Trained Linear Regression model

7.Evaluated the model using MAE and R^2 score

8.Predicted house prices

Sorce Code :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
import pandas as pd

data = pd.read_csv('/content/train.csv')
data.head()
```

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeatur
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	Nal
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	Nal
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	Nal
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	Nal
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	Nal

5 rows × 81 columns

```
data = pd.read_csv('train.csv')

# Display first 5 rows
data.head()
```

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeatur
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	Nal
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN	Nal
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	Nal
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	Nal
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN	Nal

5 rows × 81 columns

```
features = ['GrLivArea', 'BedroomAbvGr', 'FullBath']
X = data[features]
y = data['SalePrice']
```

```
X = X.fillna(X.mean())
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
print("Mean Absolute Error:", mean_absolute_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
Mean Absolute Error: 35788.061292436294
R2 Score: 0.6341189942328371
```

```
print("Intercept:", model.intercept_)

coeff_df = pd.DataFrame(model.coef_, features, columns=['Coefficient'])
print(coeff_df)
```

```
Intercept: 52261.74862694461
              Coefficient
GrLivArea       104.026307
BedroomAbvGr -26655.165357
FullBath      30014.324109
```

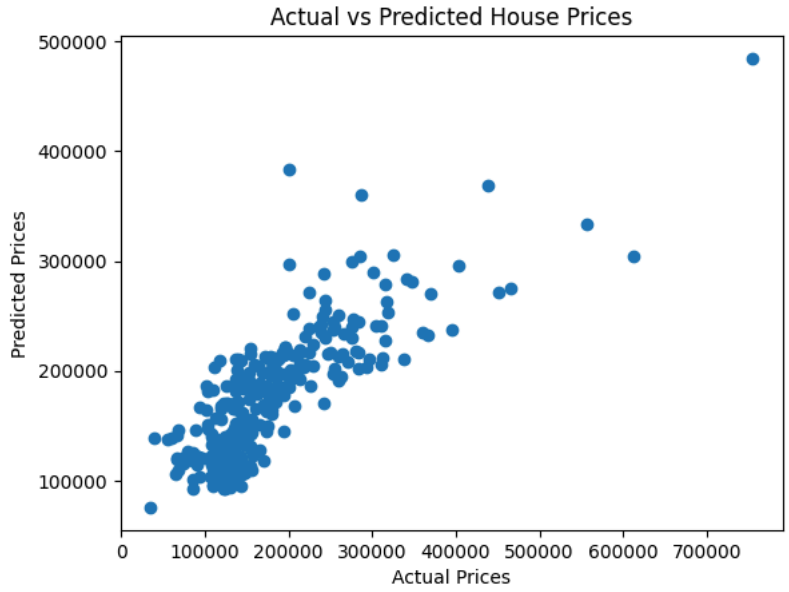
```
# Example: 1800 sq ft, 3 bedrooms, 2 bathrooms
sample = np.array([[1800, 3, 2]])

predicted_price = model.predict(sample)
print("Predicted House Price:", predicted_price[0])
```

```
Predicted House Price: 219572.25339490993
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegr
  warnings.warn(
```

```
print("MAE:",mean_absolute_error(y_test, y_pred))
print("Accuracy:", r2_score(y_test, y_pred) * 100, "%")
```

```
MAE: 35788.061292436294
Accuracy: 63.411899423283714 %
```

```
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices")
plt.show()
```



Result : The model was successfully trained and tested. It predicted house prices with reasonable accuracy. The R^2 score indicates how well the model fits the data.

Conclusion : Linear Regression is effective for predicting house prices using basic features. The model can be improved by adding more features and tuning parameters.

Future Scope : The model can be improved by using advanced algorithms like Random Forest or Gradient Boosting for better accuracy.